In the triangle OLP, $OP^2 = OL^2 + LP^2$; hence $|\overrightarrow{OP}| = \sqrt{x^2 + y^2}$.

That is, $|\vec{r}| = r = \sqrt{x^2 + y^2}$.

Observe that if $\hat{i}$ and $\hat{j}$ are the unit vectors in the postive directions of x and y axes, then the position vector of the point (6,4) can be written as $6\hat{i} + 4\hat{j}$ and this is the only way of writing it.

Result 8.8

If $\vec{a}$ and $\vec{b}$ are two non-collinear vectors in a plane, then any vector in the plane can be written as the linear combination of $\vec{a}$ and $\vec{b}$ in a unique way. That is, any vector in the plane is of the form $l\vec{a} + m\vec{b}$ for some scalars l and m.

Proof

Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, and $\vec{r}$ be any vector coplanar with $\vec{a}$ and $\vec{b}$.

Draw PL parallel to OB. Clearly $\overrightarrow{LP} = m\vec{b}$ and $\overrightarrow{OL} = l\vec{a}$ for some l and m.

Now $\overrightarrow{OP} = \overrightarrow{OL} + \overrightarrow{LP}$.

That is, $\vec{r} = l\vec{a} + m\vec{b}$.

Therefore if $\vec{r}, \vec{a}, \vec{b}$ are coplanar then $\vec{r}$ is a linear combination of $\vec{a}$ and $\vec{b}$.

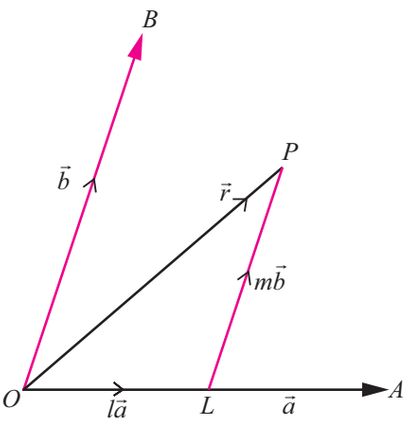


Fig. 8.30

Note 8.2

Further if three non collinear vectors are coplanar then any one of the vector can be written as a linear combination of other two. Note that the converse is also true.

Result 8.9

If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors in the space, then any vector in the space can be written as $l\vec{a} + m\vec{b} + n\vec{c}$ in a unique way for some scalars l, m and n.

Definition 8.15

Let $\hat{i}$ and $\hat{j}$ be the unit vectors in the positive directions of x and y axes respectively. Let $\vec{r}$ be any vector in the plane. Then $\vec{r} = x\hat{i} + y\hat{j}$ for some real numbers x and y. Here $x\hat{i}$ and $y\hat{j}$ are called the **rectangular components** of $\vec{r}$ along the x and y axes respectively in two dimension.

What we discussed so far can be discussed in the three dimensional space also.

8.6.2 Resolution of a vector in three dimension

Theorem 8.6

Let $\hat{i}$, $\hat{j}$ and $\hat{k}$ be the unit vectors in the direction of postive x, y and z axes respectively having initial point at the origin O. Let $\overrightarrow{OP}$ be the position vector of any point P in the space. Then $\overrightarrow{OP}$ can be uniquely written as $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$ for some real numbers x, y and z. Further $|\overrightarrow{OP}| = \sqrt{x^2 + y^2 + z^2}$.

Proof

Let (x, y, z) be the coordinates of the point P. Let Q be the foot of the perpendicular drawn from P to the xy-plane. Let R and S be the foots of the perpendiculars drawn from Q to the x and y axes respectively. Let $\overrightarrow{OP} = \vec{r}$.

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